

Aarthi Padmanabhan

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SUMMARY

Detail-oriented Data Scientist with 2 years of research experience in predictive modeling, time-series forecasting, Generative AI and cloud-based ML deployment. 2 years of industry experience as former Software Developer I at Oracle with expertise in Python, SQL, AWS.

EDUCATION

Michigan State University - East Lansing, MI	August 2023 – May 2025
Master of Science, Data Science	GPA: 3.95/4.0
SASTRA Deemed University - Thanjavur, Tamil Nadu, India	July 2017 – August 2021
Bachelor of Technology, Computer Science and Engineering	GPA: 8.31/10.0

WORK EXPERIENCE

Michigan State University College of Engineering (Alocilja Nano-Biosensors lab)	East Lansing, MI
Data Scientist	June 2025 – Present
• Built a Flask-based web application to classify 6 bacterial species using a neural network model; performed hyperparameter tuning (best val. accuracy: 87.7%), reduced input features from 32 to 12, and deployed the MLP model with 88% test accuracy enabling rapid, bacterial identification from microscopy data to support early-stage drug discovery workflows. • Building a local LLM-powered Q&A system using LangChain, OpenAI GPT-3.5-turbo-16k, ChromaDB, and Flask to extract answers from biosensor research PDFs, aiming to streamline literature review by enabling researchers to query dense scientific documents through natural language. • Developing a cross-platform Flutter app to capture and analyze biosensor images by sampling brightness values offering a low-cost, portable alternative to spectrometers for preliminary lab diagnostics and visual analysis.	
Jackson National Life	East Lansing, MI
Data Science intern (Capstone)	August 2024 – December 2024
• Explored and preprocessed a structured dataset with 97K records from 7 companies and 55 unique RILA (Registered Index Linked Annuity) products (2018 - 2024), segmented by product type (accumulation/income), fee structure, term length, protection options, and index types. • Engineered training datasets using Sliding Window and Oversampling methods to address class imbalance in OHIT (Over sampled High dimensional Imbalanced Time series), improving minority class recall by 20%. • Developed and evaluated LSTM and GRU models for cap rate classification; GRU outperformed LSTM with a 5% boost in validation accuracy and stronger precision and recall across multiple products. • Forecasted future cap rates using SARIMAX, capturing seasonal patterns with less than 3% prediction deviation, enabling data driven planning for rate adjustments and strategic pricing decisions.	
Bruno Basso Lab MSU	East Lansing, MI
Data Science intern	May 2024 – August 2024
• Collected and processed 32K geospatial data points using QGIS and Google Earth Engine API with Python, extracting 12 spectral bands from Landsat and Sentinel-2 imagery. • Reduced data collection time by 70% through parallelized cloud-based satellite data processing. • Engineered 2 vegetation indices (NDVI and EVI) as predictive features by applying satellite-specific band combinations tailored to satellite type and temporal range (1982–2022). • Improved data quality by 25% via satellite-specific cloud masking, spatial and temporal filtering, and median aggregation, minimizing noise and enhancing vegetation index reliability. • Tuned a Random Forest Regressor model to predict N₂O emissions, capturing 52% of the variability in the test set and identifying EVI as the dominant predictor, contributing over 60% to the model's feature importance. • Identified 180 data points with N₂O emissions exceeding 100 gN/ha/day, highlighting critical emission hotspots and supporting targeted nitrogen management strategies to promote sustainable agriculture.	
Oracle	Bengaluru, Karnataka, India
Software Developer I	August 2021 – August 2023
• Designed, developed, and analyzed functional tests for MySQL, ensuring robust performance across on-premise deployments and the Heatwave Cloud Service (HCS). • Automated 2 core MySQL enterprise features (audit and firewall), reducing manual testing effort by 80–90% and enhancing cloud deployment reliability using Java. • Resolved bugs in QA test tools, leading to a 25% improvement in tool stability for developers and QA teams using Perl. • Participated in weekly monitoring of test runs and contributed to the filing of critical bugs every 2 months, ensuring timely issue identification and resolution. • Supported code reviews and MySQL Database Service operations, helping resolve 10+ high-severity customer tickets per quarter, maintaining SLA compliance and production reliability.	

PROJECTS

Research Assistant – Multi-Modal RAG System with LLMs & LanceDB

- Built a document-aware Q&A assistant using RAG, supporting PDF upload, querying, and summarization via Gradio UI.
- Processed text and image chunks with PyMuPDF and nomic-ai/sentence-transformers, stored in LanceDB, and queried using LLaMA-Vision.
- Deployed via Docker & AWS, with evaluation tracked using Evidently (ROUGE-1: 0.024, Recall@k: 0.083).

Cell Image Classification – Deep Learning for Biomedical Imaging

- Processed bright-field cell images (THP1, MCF7, PBMC) using normalization, resizing, and augmentation (flip, blur, noise) to enhance data quality.
- Implemented and compared SVM (RBF kernel) and CellNet, a CNN architecture designed for robust biomedical image classification.
- CellNet achieved 92% accuracy, outperforming SVM by 7% on noisy datasets.

StoreCast – Time Series Forecasting, MLOps, Monitoring & Deployment

- Developed a forecasting platform for Walmart store sales using Auto ARIMA and XGBoost, aggregating historical weekly sales from 45 stores and storing models and data in AWS S3.
- Exposed predictions via a Flask API deployed on AWS EC2 using Docker, with monitoring and metrics collection through Prometheus and Grafana dashboards for forecast latency.
- Enabled dynamic model loading, ensemble forecasting, and real-time performance tracking, demonstrating production-grade MLOps integration with 78% ensemble accuracy and 9.42% MAPE.

SKILLS

Programming Languages: Python (Pandas, NumPy), Java, R, C++, SQL

Database Technologies: MySQL, MongoDB

Data Engineering & Visualization: ETL Pipelines, Matplotlib, Seaborn, Plotly, Tableau, Power BI

Machine Learning & Statistical Analysis: scikit-learn, Hypothesis Testing, A/B Testing, statsmodels, spaCy

Deep Learning & Computer Vision: TensorFlow, PyTorch, OpenCV, HuggingFace

Cloud Service: Oracle Cloud Infrastructure (OCI), Amazon Web Services (AWS) – EC2, S3, SageMaker

DevOps & Deployment Tools: Version Control (Git, GitHub), CI/CD (Jenkins), Containerization (Docker), Monitoring (Prometheus, Grafana), Web Frameworks (Flask API, Streamlit), Linux CLI

Other Tools: Jupyter Notebook, QGIS, Jira, Confluence

LEADERSHIP & ENGAGEMENT

Led 33 recitation sessions as a Graduate Teaching Assistant for STT 231: Statistics for Scientists course at Michigan State University, mentoring 45 undergraduate students through interactive instruction in R, multivariate analysis, hypothesis testing, and real-world statistical applications; provided individualized support via weekly office hours, grading, and exam proctoring.